THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — MATHS · HARD LEVEL CHAPTER TEST SET 2

Chapter 6: **Number Play** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Kaprekar's routine on 6321 reaches 6174 after how many steps? 1

(a) 8 **(b)** 9 **(c)** 7 **(d)** 6

2. The Collatz sequence from 15 needs how many steps to reach 1? 1

(a) 18 **(b)** 16 **(c)** 17 **(d)** 19

3. For every whole number n, $n + (n+1) + (n+2)$ is always a multiple of: 1

(a) 3 **(b)** 4 **(c)** 6 **(d)** 2

4. If $1 + 2 + 3 + \cdots + n = 406$, then n is: 1

(a) 29 **(b)** 28 **(c)** 30 **(d)** 27

5. The number of reverse-and-add steps for 79 to become a palindrome is: 1

(a) 5 **(b)** 6 **(c)** 7 **(d)** 8

6. Which expression is a multiple of 11 for every digit pair (a, b)? 1

(a) $10a + b$ **(b)** $(10a + b) - (10b + a)$ **(c)** $(10a + b) + (10b + a)$ **(d)** $a - b$

7. The smallest 5-digit number divisible by 9 is: 1

(a) 10008 **(b)** 10017 **(c)** 10007 **(d)** 10009

8. The largest 4-digit number with all digits different that is divisible by 10 is: 1

(a) 9872 **(b)** 9869 **(c)** 9870 **(d)** 9871

9. Kaprekar's routine on 1459 reaches 6174 after how many steps? 1

(a) 5 **(b)** 3 **(c)** 4 **(d)** 2

10. The Collatz sequence from 17 needs how many steps to reach 1? 1

(a) 13 **(b)** 11 **(c)** 12 **(d)** 14

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Show that $n(n+1)$ is even, and verify for $n = 24$. 2

12. Using each of the digits $6, 4, 1, 0$ once, make the largest 4-digit number divisible by 5. 2

13. A two-digit number added to its reverse gives 121 and subtracted from it gives 63. Find the number. 2

14. Show that $n(n+1)$ is even, and verify for $n = 37$. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. The number $5 * 8 * 4$ has the same digit $*$ in both blanks and is divisible by 9. Find all possible values of $*$. 3

16. Start with 1279 and apply Kaprekar's routine until 6174 appears. Show every step. 3

17. In how many ways can 27 be written as the sum of two or more consecutive positive whole numbers? List them. 3

OR

The number $3 * 6 * 9$ has the same digit $*$ in both blanks and is divisible by 9. Find all possible values of $*$. 3

18. Start with 1327 and apply Kaprekar's routine until 6174 appears. Show every step. 3

OR

In how many ways can 33 be written as the sum of two or more consecutive positive whole numbers? List them. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Use reverse-and-add on 164 until a palindrome is formed. Show each step, state the number of steps and the final palindrome. 5

OR

Find all two-digit numbers that are exactly 4 times the sum of their digits. 5

20. Use reverse-and-add on 174 until a palindrome is formed. Show each step, state the number of steps and the final palindrome. 5

OR

Find all two-digit numbers that are exactly 5 times the sum of their digits. 5

END OF QUESTION PAPER — Total: 40 Marks

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